NAVAL ARCHITECTURE
TRIM

1 A ship 160 m long has draughts 7.70 m forward and 8.25 m aft. MCT1cm is 250 tonne.m and LCF is 1.8 m forward of amidships. Calculate the new draughts after a mass of 200 tonne was moved longitudinally aft a distance of 55 m.

[Ans: 7.68 m; 8,27 m]

2 A ship of length 122 m has the centre of flotation 7.62 m abaft amidships. In a particular condition of loading the measured draughts are 6.56 m and 7.60 m forward and aft respectively. Determine the true mean draught.

[Ans: 7.145 m]

3 A ship of length 107 m has draughts 6.1 m fwd and 5.5 m aft. Determine the true mean draught if LCF = 3 m fwd of amidships.

[Ans: 5.817 m]

4 A ship of length 107 m has a displacement of 6330 tonnes when floating at draughts 6.1 m fwd and 5.5 m aft. KM_L = 140 m and KG = 4.5 m. If LCF = 3 m fwd of amidships. Determine the new end draughts when 100 tonnes of cargo is moved forward a distance of 30 m.

[Ans: 6.277 m; 5.302 m]

5 A ship of length 100 m has a displacement of 6350 tonnes when floating at draughts 5.5 m fwd and 6.1 m aft. KM_L = 140 m and KG = 5 m. If LCF = 3 m fwd of amidships. Determine the distance that 100 tonnes should be moved to bring the vessel on an even keel.

[Ans: 51.46 m F]

6 A ship of length 110 m has a displacement of 6300 tonnes when floating at draughts 6.1 m fwd and 6.5 m aft. Determine the effect on the draughts of filling a double bottom tank whose capacity is 216 tonnes and has a centroid of tank 18 m forward of the centre of flotation of the vessel. Assume that the GM_L = 136 m, LCF = 3.5 m fwd of midship and TPC = 12.

[Ans: 6.544 m; 5.447 m]

7 A ship of length 107 m has a displacement of 6330 tonnes when floating at draughts 6.1 m fwd and 5.5 m aft. Determine the end draughts when 204 tonnes of fuel is consumed at a centroid 9 m fwd of the centre of flotation, the LCF is 3.5 m aft of midships. MCT1cm = 79.5 tonne.m and TPC = 12.

[Ans: 5.807 m; 5.438 m]

8 A ship of 250 m length arrives at a loading berth displacing 28 000 tonne. The centre of gravity is 7.0 m above the keel and 2.0 m aft of midships.

The following changes now take place:

- 8000 tonne of water ballast are discharged from Kg 5.0 m and Lcg 50.0 m forward of midships.
- 1500 tonne fuel is taken on board at Kg 2.0 m and Lcg 88.8 m aft of midships.
- 50 270 tonne of cargo is loaded at Kg 8.0 m and Lcg 16.0 m forward of midships.

The following particulars have been taken from the ships hydrostatic table:

d (m)	Δ (Tonne)	KM_L (m)	LCB (m)	LCF (m)
12.0	66 595	357.1	5.02 F	2.90 A
12.6	77 845	369.7	4.78 F	3.38 A

Determine the draughts for the final condition of loading.

[11.59 m; 12.93 m]

9 A ship 90 m long displaces 5200 tonne and floats at draughts of 4.95 m forward and 5.35 m aft when in sea water of 1023 kg/m^3. The waterplane area is 1100 m^2, GM_L 95.0 m, LCB 0.6 m forward of midships and LCF 2.2 m aft of midships.

Calculate the new end draughts when the vessel moves into fresh water of 1002 kg/m^3.

[5.078 m; 5.419 m]

10 A ship of length 69 m floats in fresh water 0.996 m^3/tonne with draughts forward and aft of 4.25 m and 5.03 m respectively. Calculate the displacement in tonnes using the following hydrostatic information for salt water 0.976 m^3/tonne.

Displacement at 4.65 m mean draught = 2352 tonne
Corresponding LCF = 1.98 m fwd of midships
Corresponding TPC = 5.8

What is the change in trim when she now passes into salt water, assuming that the MCT1cm is 62 tonne.m and LCB is 1.37 m forward of midships, and what will be the new end draughts?

[Ans: 4.165 m; 4.95 m]

11 A ship of length 135 m floats in sea water 1025 kg/m^3. In the light condition the ship has a displacement 5000 tonne and the centre of gravity is at midships. Loading now takes place as detailed in table below:

LOAD	MASS (Tonne)	Lcg from midhip (m)
General cargo	3890	40.0 F
General cargo	3380	34.0 A
Refrigerated cargo	680	48.0 A
Oil fuel	720	15.7 F
Fresh water	80	58.0 F
Stores, etc.	50	33.6 A

Determine the draughts forward and aft for the loaded condition using the relevant data extracted from the accompanying hydrostatic curves.

[7.37 m; 7.83 m]

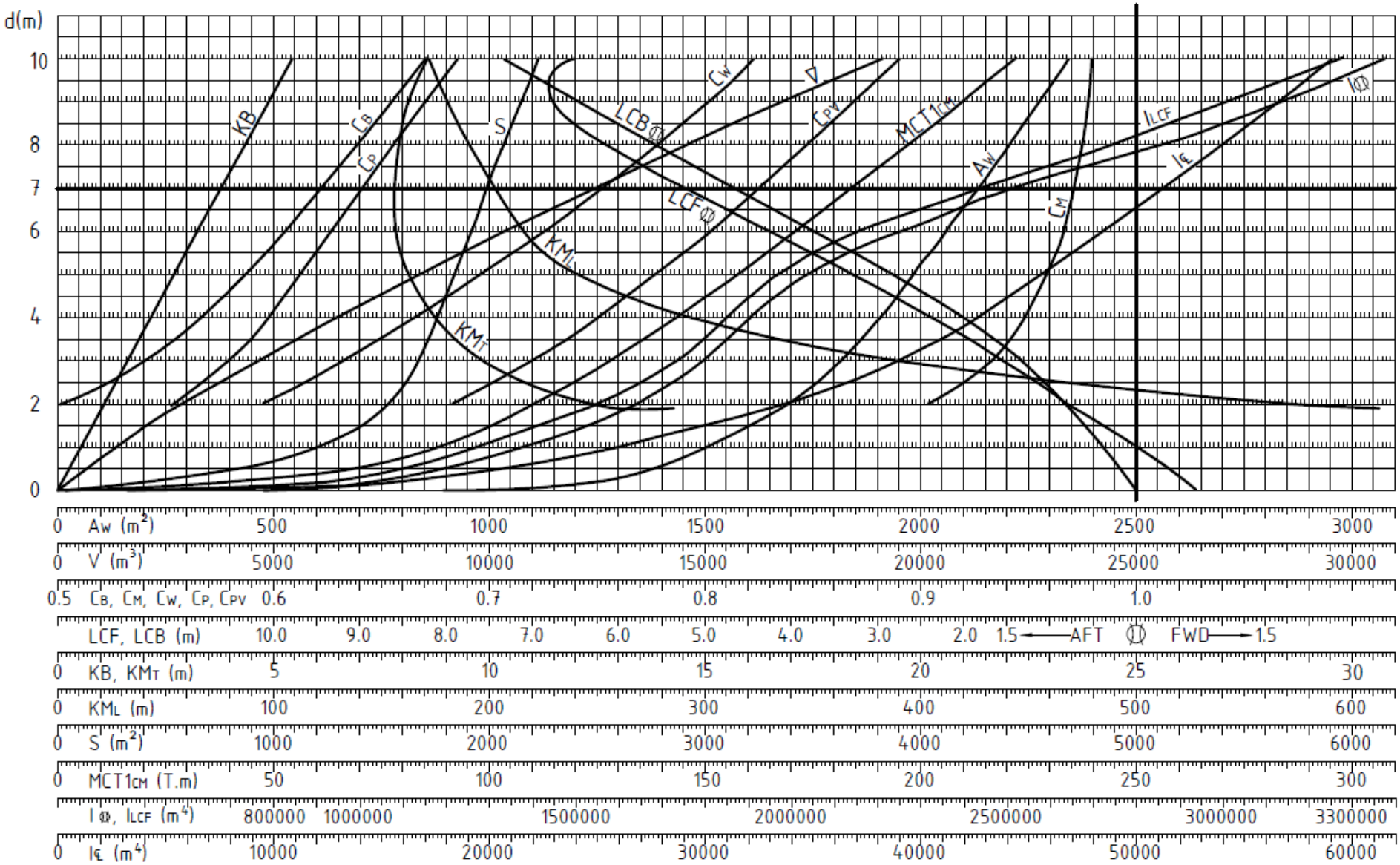


Figure 1 - HYDROSTATIC CURVES

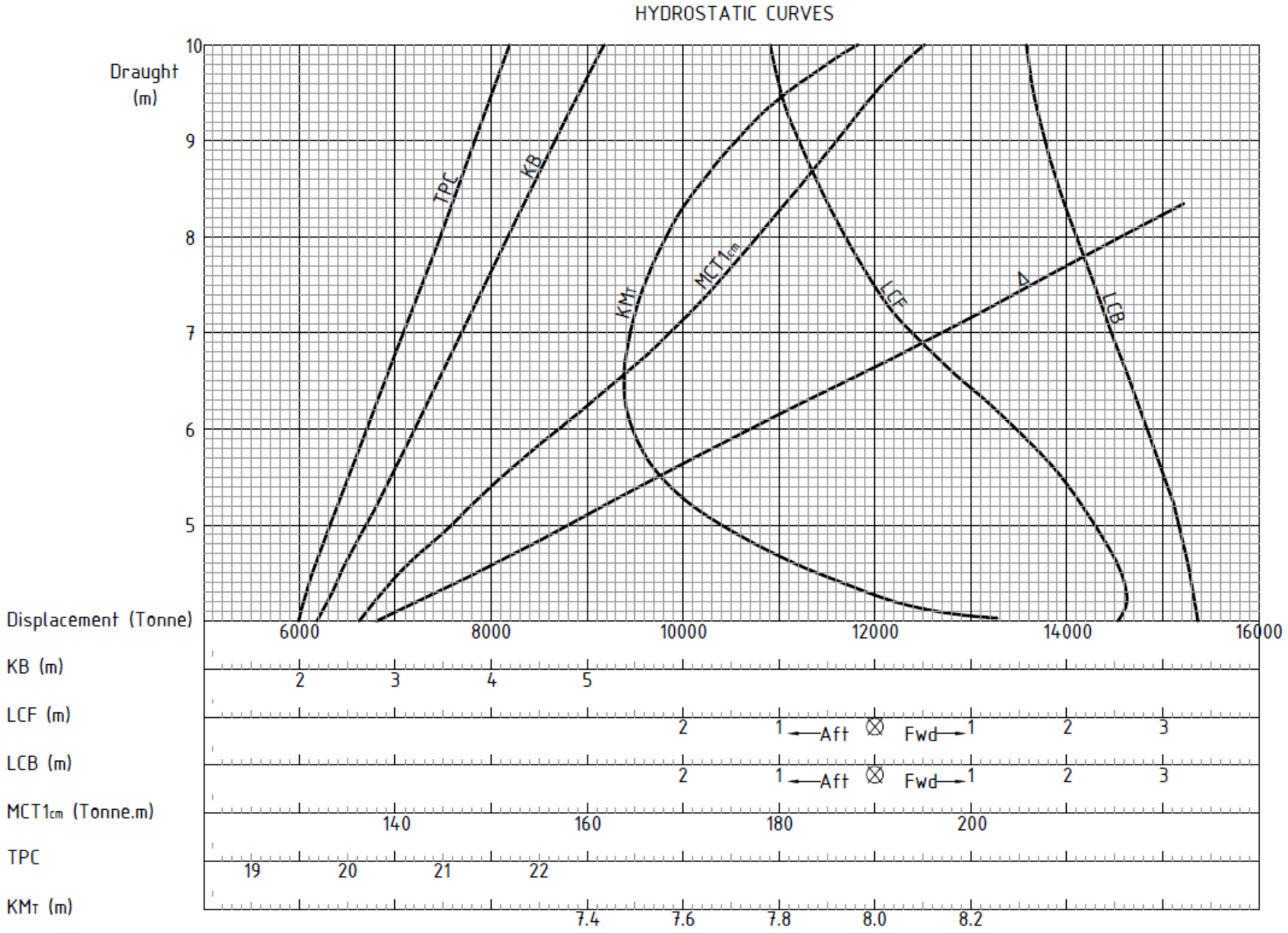


Figure 2 - HYDROSTATIC CURVES